Scientific Citation Retrieval

NDCG@10: 0.45 $\rightarrow$ 0.67

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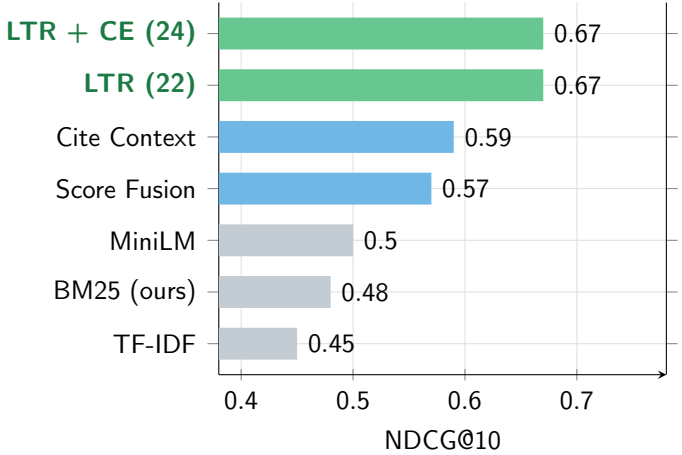
Results - Overview



Baselines

Our approach

Learning to Rank

Scripts 33 & 34
reranking < 0.67 $\downarrow$

Baselines

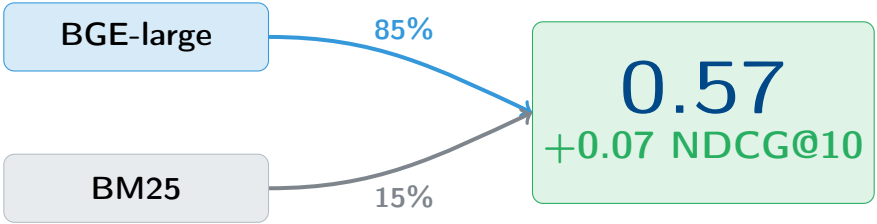
TF-IDF
(01) - provided
0.45
NDCG@10

Dense MiniLM
(02) - provided
0.50
NDCG@10

BM25
(03) - ours
0.48
NDCG@10

Ceiling: sparse retrieval only → exact term matching

Score Fusion (Script 10)



scores min-max normalized, weights tuned by grid search

Citation Context Mining - Script 21

Full text completely ignored until now

"As shown by [1], neural networks can..."

"(Smith et al., 2020) demonstrated that..."

1. BM25 index on corpus **full text**
2. Extract citation sentences via `regex`
3. Score against full-text index
4. Fuse via 2D grid search

0.59

+0.02 NDCG@10

still sparse and still plateauing

Learning to Rank - Architecture (Script 22)

∼16 features per (query, doc) pair:

- Dense scores: UAE, BGE, E5, SciNCL
- Sparse scores: BM25 TA, BM25 fulltext,
  cite-ctx BM25, TF-IDF
- Reciprocal rank per system
- Year proximity
- Domain match

XGBRanker

`objective: rank:ndcg`
`max_depth = 4`
`n_estimators = 200`

candidates: top-200 per system
eval: 5-fold GroupKFold

Learning to Rank - Results

0.67

+0.08 NDCG@10

biggest jump in the project

Script 24 - adding cross-encoder features:
3 new features: `ce_score`, `has_ce_score`, `ce_rank`

Reranking - Scripts 33 & 34

CE Reranking (33)

bge-reranker-v2-m3 on top-50

Blend: CE score + LTR rank

$<0.67 \downarrow$

CE already baked into LTR (script 24)
$\rightarrow$ no new information

LLM Reranking (34)

Qwen2.5-7B rates 0–10 per pair

top-20 only (cost)

$<0.67 \downarrow$

7B model too imprecise at this level
LTR already well-calibrated

Conclusion

NDCG@10 0.45 $\longrightarrow$ 0.67
(+49%)

1. Better signals: MiniLM $\rightarrow$ BGE-large **+0.07**
2. Learning signal interactions (LTR) **+0.08**
3. Reranking on LTR $\rightarrow$ regression $\downarrow$